NEET UG 2027: Master Chapter Theory

Electrodynamics, Magnetism, Electromagnetic Induction & AC Circuits

Prepared, Compiled & Authored by Satya Prakash · Strictly Calibrated to Revised NMC Syllabus

1 Electrostatics, Coulomb's Law & Continuous Charge Distributions

Foundations of Electric Charge & Coulomb's Law in Vector Form

1. Fundamental Charge Properties:

- **Quantization of Charge:** Charge exists strictly as integer multiples of the elementary charge $e \approx 1.602 \times 10^{-19}\,\text{C}$:

$$q = \pm ne \quad (n \in \mathbb{N}) \tag{1}$$

- **Conservation of Charge:** The algebraic sum of fundamental charges within an isolated physical system remains invariant over time.
- **Invariance of Charge:** The magnitude of electric charge is independent of the reference frame speed ($q_{\text{rest}} = q_{\text{motion}}$), unlike relativistic mass.

2. Coulomb's Inverse-Square Law in Vector Form: The electrostatic force exerted by point charge q_1 at position vector $\mathbf{r}_1$ on charge q_2 at $\mathbf{r}_2$:

$$\mathbf{F}_{12} = \frac{1}{4\pi\varepsilon_0\varepsilon_r}\frac{q_1 q_2}{|\mathbf{r}_2 - \mathbf{r}_1|^3}(\mathbf{r}_2 - \mathbf{r}_1) = \frac{1}{4\pi\varepsilon_0\varepsilon_r}\frac{q_1 q_2}{r_{12}^2}\hat{\mathbf{r}}_{12} \tag{2}$$

where $\varepsilon_0 = 8.854 \times 10^{-12}\,\text{C}^2 \cdot \text{N}^{-1} \cdot \text{m}^{-2}$ is the permittivity of free space, and $\frac{1}{4\pi\varepsilon_0} \approx 8.988 \times 10^9\,\text{N} \cdot \text{m}^2 \cdot \text{C}^{-2}$. In a dielectric medium of relative permittivity (dielectric constant) $\kappa = \varepsilon_r > 1$:

$$F_{\text{medium}} = \frac{F_{\text{vacuum}}}{\kappa} \tag{3}$$

3. Superposition Principle: The resultant force acting on any test charge q_0 due to a collection of discrete charges $q_1, q_2, \ldots, q_n$ is the vector sum:

$$\mathbf{F}_{\text{net}} = \sum_{i=1}^{n}\mathbf{F}_{i0} = \frac{q_0}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{|\mathbf{r}_0 - \mathbf{r}_i|^3}(\mathbf{r}_0 - \mathbf{r}_i) \tag{4}$$

Electric Field & Potential of Continuous Charge Geometries: Ring & Disk

1. Electric Field Definition: $\mathbf{E} = \lim_{q_0 \to 0}\frac{\mathbf{F}}{q_0} = -\boldsymbol{\nabla} V$. **2. Uniformly Charged Thin Circular Ring on its Symmetry Axis:** Consider a ring of total charge Q and radius R positioned in the $y-z$ plane with symmetry axis along x.

- By radial symmetry, perpendicular field components cancel: $dE_\perp = 0$.
- Axial field element: $dE_x = dE\cos\theta = \frac{1}{4\pi\varepsilon_0}\frac{dq}{x^2+R^2}\frac{x}{\sqrt{x^2+R^2}}$.
- Total integrated axial electric field:

$$\mathbf{E}(\mathbf{x}) = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}}\frac{\mathbf{Qx}}{(\mathbf{x^2 + R^2})^{\mathbf{3/2}}}\hat{\mathbf{i}} \tag{5}$$

- **At Ring Center ($x = 0$):** $E(0) = 0$.
- **Far-Field Regime ($x \gg R$):** $E(x) \approx \frac{1}{4\pi\varepsilon_0}\frac{Q}{x^2}$ (acts as point charge).
- **Location of Maximum Electric Field:**

$$\frac{dE}{dx} = \frac{Q}{4\pi\varepsilon_0}\frac{(x^2+R^2)^{3/2} - x\cdot\frac{3}{2}(x^2+R^2)^{1/2}(2x)}{(x^2+R^2)^3} = 0 \implies x^2 + R^2 - 3x^2 = 0 \implies \mathbf{x} = \pm\frac{\mathbf{R}}{\sqrt{\mathbf{2}}} \tag{6}$$

$$\mathbf{E}_{\max} = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{2Q}}{\mathbf{3\sqrt{3}R^2}} \tag{7}$$

- **Electrostatic Potential on Ring Axis:**

$$\mathbf{V(x)} = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{Q}}{\sqrt{\mathbf{x^2 + R^2}}} \implies V_{\text{center}} = \frac{Q}{4\pi\varepsilon_0 R} \tag{8}$$

3. Uniformly Charged Flat Circular Disk on its Axis: Integrating concentric rings of radius r and charge $dq = \sigma(2\pi r dr)$ from $r = 0$ to R:

$$\mathbf{E(x)} = \frac{\sigma}{\mathbf{2\varepsilon_0}} \left(\mathbf{1} - \frac{\mathbf{x}}{\sqrt{\mathbf{x^2 + R^2}}}\right) \hat{\mathbf{i}}, \qquad \mathbf{V(x)} = \frac{\sigma}{\mathbf{2\varepsilon_0}} \left(\sqrt{\mathbf{x^2 + R^2}} - \mathbf{x}\right) \tag{9}$$

For $R \to \infty$ (infinite sheet), $E = \frac{\sigma}{2\varepsilon_0}$ (constant, independent of x).

Electric Dipole Dynamics: Field, Torque & Potential Energy

An electric dipole consists of two equal and opposite point charges $+q$ and $-q$ separated by a displacement vector $2\mathbf{a}$ pointing from $-q$ to $+q$:

$$\mathbf{p} = q(2\mathbf{a}) \quad [\text{C} \cdot \text{m}] \tag{10}$$

1. Electric Field of Dipole in Coordinate Space ($r \gg a$):

- **Axial Point (End-on Position, $\theta = 0°$):**

$$\mathbf{E}_{\text{axial}} = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{2p}}{\mathbf{r^3}} \quad \text{(Collinear and parallel to } \mathbf{p}\text{)} \tag{11}$$

- **Equatorial Point (Broadside-on Position, $\theta = 90°$):**

$$\mathbf{E}_{\text{eq}} = -\frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{p}}{\mathbf{r^3}} \quad \text{(Antiparallel to } \mathbf{p}\text{)} \tag{12}$$

- **General Coordinate Point (r, θ):**

$$|\mathbf{E}| = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{p}}{\mathbf{r^3}} \sqrt{\mathbf{1} + \mathbf{3}\cos^2\theta}, \qquad \tan\alpha = \frac{1}{2}\tan\theta \tag{13}$$

  where α is the angle between $\mathbf{E}$ and position vector $\mathbf{r}$.

- **Electrostatic Potential at (r, θ):**

$$\mathbf{V}(\mathbf{r}, \theta) = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{\mathbf{r^2}} = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{p}\cos\theta}{\mathbf{r^2}} \tag{14}$$

  Notice: $V_{\text{eq}} = V(r, 90°) = 0$ everywhere along the equatorial bisector plane.

2. External Uniform Electric Field Interactions:

- Net Force: $\mathbf{F}_{\text{net}} = \mathbf{qE} + (-\mathbf{q})\mathbf{E} = \mathbf{0}$ (zero translational acceleration).
- Torque:

$$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E} \implies |\boldsymbol{\tau}| = pE\sin\theta \tag{15}$$

  Maximum torque occurs at $\theta = 90°$ ($\tau_{\max} = pE$).

- Potential Energy of Dipole in External Field:

$$\mathbf{U} = -\mathbf{p} \cdot \mathbf{E} = -\mathbf{pE}\cos\theta \tag{16}$$

  - **Stable Equilibrium ($\theta = 0°$):** $\mathbf{p} \parallel \mathbf{E} \implies \boldsymbol{\tau} = 0$, $U_{\min} = -pE$.
  - **Unstable Equilibrium ($\theta = 180°$):** $\mathbf{p} \parallel -\mathbf{E} \implies \boldsymbol{\tau} = 0$, $U_{\max} = +pE$.

- Work done by external agent in rotating dipole from θ_1 to θ_2:

$$\mathbf{W}_{\text{ext}} = \mathbf{\Delta U} = \mathbf{pE}(\cos\theta_1 - \cos\theta_2) \tag{17}$$

2 Gauss's Law, Electric Flux & Boundary Value Applications

Gauss's Law Formulation & Electric Flux

1. Electric Flux (Φ_E): Surface integral of the electric field vector through an oriented surface element $d\mathbf{A} = \hat{\mathbf{n}} dA$:

$$\Phi_E = \iint_S \mathbf{E} \cdot d\mathbf{A} \quad [\mathrm{N \cdot m^2 \cdot C^{-1}} \text{ or } \mathrm{V \cdot m}] \tag{18}$$

2. Gauss's Law (Maxwell's First Equation): The net outward electric flux through any closed Gaussian surface enclosing net charge q_{enclosed}:

$$\oint_{\partial \mathbf{V}} \mathbf{E} \cdot \mathbf{dA} = \frac{\mathbf{q}_{\text{enclosed}}}{\varepsilon_0} \tag{19}$$

Key Physical Insights:

- Net flux is completely independent of the shape, size, or geometry of the Gaussian boundary.
- Charges residing outside the closed surface contribute zero net flux through the surface (entering flux equals exiting flux).
- Electric field $\mathbf{E}$ on the left-hand side is the **net resultant field** produced by all charges (both inside and outside).

Master Applications of Gauss's Law: Exact Field Derivations

1. Infinitely Long Line Charge of Uniform Density λ (C/m): Using a coaxial Gaussian cylinder of radius r and length L:

$$\oint \mathbf{E} \cdot d\mathbf{A} = E(2\pi r L) = \frac{\lambda L}{\varepsilon_0} \implies \mathbf{E}(\mathbf{r}) = \frac{\lambda}{\mathbf{2\pi\varepsilon_0 r}} \hat{\mathbf{r}} \tag{20}$$

2. Uniformly Charged Infinite Thin Flat Sheet of Density σ (C/m^2): Using a Gaussian pillbox of cross-sectional area A piercing normal to the sheet:

$$\oint \mathbf{E} \cdot d\mathbf{A} = 2EA = \frac{\sigma A}{\varepsilon_0} \implies \mathbf{E} = \frac{\sigma}{\mathbf{2\varepsilon_0}} \hat{\mathbf{n}} \quad \text{(Independent of distance)} \tag{21}$$

3. Surface of a Charged Conductor: Inside the conductor, $E = 0$. Flux exits only from the outer face:

$$EA = \frac{\sigma A}{\varepsilon_0} \implies \mathbf{E}_{\text{conductor}} = \frac{\sigma}{\boldsymbol{\varepsilon_0}} \hat{\mathbf{n}} \quad \text{(Twice that of a non-conducting sheet)} \tag{22}$$

4. Thin Spherical Shell of Radius R and Total Charge Q:

- **Interior Region ($r < R$):** Gaussian surface encloses $q_{\text{encl}} = 0 \implies \mathbf{E}_{\text{in}} = \mathbf{0}$.
- Potential is constant throughout interior: $\mathbf{V}_{\text{in}} = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{Q}}{\mathbf{R}}$.
- **Exterior Region ($r \geq R$):** Gaussian sphere encloses $Q \implies \mathbf{E}_{\text{out}}(\mathbf{r}) = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{Q}}{\mathbf{r^2}} \hat{\mathbf{r}}$.
- Exterior potential: $\mathbf{V}_{\text{out}}(\mathbf{r}) = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{Q}}{\mathbf{r}}$.

5. Uniformly Charged Non-Conducting Solid Sphere of Radius R and Total Charge Q:

- Volume charge density: $\rho = \frac{Q}{\frac{4}{3}\pi R^3}$.
- **Interior Region ($r \leq R$):** $q_{\text{encl}} = \rho\left(\frac{4}{3}\pi r^3\right) = Q\frac{r^3}{R^3}$:

$$E(r) \cdot 4\pi r^2 = \frac{Q r^3}{\varepsilon_0 R^3} \implies \mathbf{E}_{\text{in}}(\mathbf{r}) = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{Qr}}{\mathbf{R^3}} \hat{\mathbf{r}} = \frac{\rho \mathbf{r}}{\mathbf{3\varepsilon_0}} \hat{\mathbf{r}} \tag{23}$$

  Electric field increases linearly with radial distance r from center.

- **Interior Potential:** $V_{\text{in}}(r) = -\int_{\infty}^{R} E_{\text{out}} dr - \int_{R}^{r} E_{\text{in}} dr$:

$$\mathbf{V}_{\text{in}}(\mathbf{r}) = \frac{\mathbf{Q}}{\mathbf{4\pi\varepsilon_0}} \left[\frac{\mathbf{3R^2 - r^2}}{\mathbf{2R^3}}\right] \implies \mathbf{V}_{\text{center}} = \frac{\mathbf{3}}{\mathbf{2}} \mathbf{V}_{\text{surface}} = \mathbf{1.5 V}_{\text{surface}} \tag{24}$$

- **Exterior Region ($r \geq R$):** Identical to point charge:

$$\mathbf{E}_{\text{out}}(\mathbf{r}) = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{Q}}{\mathbf{r^2}} \hat{\mathbf{r}}, \qquad \mathbf{V}_{\text{out}}(\mathbf{r}) = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{Q}}{\mathbf{r}} \tag{25}$$

NCERT Extraction: Electrostatic Shielding & Cavity Boundary Properties

Inside any hollow conducting cavity of arbitrary shape, the electric field is **identically zero** ($\mathbf{E} = 0$), provided no charge is enclosed within the cavity. The potential throughout the cavity remains constant and strictly equals the potential on the conducting surface ($V_{\text{cavity}} = V_{\text{surface}}$).

- *Faraday Cage Principle:* Any sensitive electrical apparatus enclosed in a hollow metallic enclosure is completely shielded from external electrical disturbances.
- *Practical Safety:* During lightning storms, it is significantly safer to remain inside a closed metallic car or bus than under a tree or in the open, as lightning charges flow along the outer conducting body directly to the ground without penetrating the passenger interior.

Syllabus Boundary Alert: Van de Graaff Generator Deletion

NMC REVISION WARNING FOR NEET UG: Under the revised NEET UG syllabus issued by the National Medical Commission:

- **STRICTLY DELETED:** The **Van de Graaff Generator** (high voltage production via belt charge transport, corona discharge, insulating column, spherical dome collection) has been **completely eliminated**.
- **EXAM ORIENTATION:** Candidates must master Gauss's law boundary applications (spherical shells, solid spheres, line charges, and conductor electrostatic shielding) without studying the obsolete Van de Graaff machine.

3 Capacitance, Dielectric Physics & Energy Storage

Dielectric Polarization & Capacitance Principles

1. Capacitance Definition: The capacity of a conductor to store electric charge per unit potential rise:

$$C = \frac{Q}{V} \quad [\text{Farad}, \text{F} = \text{C} \cdot \text{V}^{-1}] \tag{26}$$

For an isolated conducting sphere of radius R: $C = 4\pi\varepsilon_0 R$. **2. Parallel Plate Capacitor in Vacuum:** Plates of area A separated by vacuum gap d with uniform field $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$ and $V = Ed$:

$$\mathbf{C_0 = \frac{\varepsilon_0 A}{d}} \tag{27}$$

3. Dielectric Material Polarization: When an insulating dielectric is inserted into an external electric field $\mathbf{E}_0$, bound molecular dipoles align, generating a macroscopic polarization vector $\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}$. This induces surface bound charge density $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$ producing an internal opposing depolarizing field $\mathbf{E}_{\text{induced}} = \frac{\sigma_b}{\varepsilon_0}$.

$$\mathbf{E}_{\text{net}} = \mathbf{E_0} - \mathbf{E}_{\text{induced}} = \frac{\mathbf{E_0}}{\kappa} \quad \text{(Field is attenuated by factor } \kappa) \tag{28}$$

where $\kappa = 1 + \chi_e = \varepsilon_r$ is the dielectric constant. **4. Bound Induced Surface Charge:**

$$E_{\text{net}} = \frac{\sigma - \sigma_b}{\varepsilon_0} = \frac{\sigma}{\kappa\varepsilon_0} \implies \sigma_b = \sigma\left(1 - \frac{1}{\kappa}\right) \implies \mathbf{q_b = q\left(1 - \frac{1}{\kappa}\right)} \tag{29}$$

Capacitor with Partial Dielectric Slab & Network Combinations

1. Partial Dielectric Slab Insertion (Thickness $t < d$): Across the vacuum gap $(d - t)$, electric field is $E_0 = \frac{Q}{\varepsilon_0 A}$. Inside the slab of thickness t, field is $E = \frac{E_0}{\kappa}$. Total potential difference between plates:

$$V = E_0(d - t) + Et = E_0\left[(d - t) + \frac{t}{\kappa}\right] = \frac{Q}{\varepsilon_0 A}\left[d - t\left(1 - \frac{1}{\kappa}\right)\right] \tag{30}$$

$$\mathbf{C = \frac{Q}{V} = \frac{\varepsilon_0 A}{d - t\left(1 - \frac{1}{\kappa}\right)}} \tag{31}$$

Special Cases:

- Complete dielectric filling ($t = d$): $C = \kappa \frac{\varepsilon_0 A}{d} = \kappa C_0$.
- Conducting slab inserted ($t < d$, $\kappa \to \infty$): $C = \frac{\varepsilon_0 A}{d-t}$.

2. Grouping of Capacitors:

- **Series Combination (Charge Q is identical):**

$$V = V_1 + V_2 + \cdots \implies \frac{\mathbf{1}}{\mathbf{C_{eq}}} = \sum_{\mathbf{i=1}}^{\mathbf{n}} \frac{\mathbf{1}}{\mathbf{C_i}} \quad \left(\text{For two: } C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2} \right) \tag{32}$$

- **Parallel Combination (Potential difference V is identical):**

$$Q = Q_1 + Q_2 + \cdots \implies \mathbf{C_{eq}} = \sum_{\mathbf{i=1}}^{\mathbf{n}} \mathbf{C_i} \tag{33}$$

3. Electrostatic Energy Stored & Field Energy Density: Work required to assemble charge Q on plates against potential V:

$$\mathbf{U} = \int_{\mathbf{0}}^{\mathbf{Q}} \mathbf{V(q)dq} = \int_{\mathbf{0}}^{\mathbf{Q}} \frac{\mathbf{q}}{\mathbf{C}} \mathbf{dq} = \frac{\mathbf{Q^2}}{\mathbf{2C}} = \frac{\mathbf{1}}{\mathbf{2}} \mathbf{CV^2} = \frac{\mathbf{1}}{\mathbf{2}} \mathbf{QV} \tag{34}$$

4. Energy Density in Electric Field: Dividing energy U by the volume of the dielectric space ($A \cdot d$):

$$u_E = \frac{U}{Ad} = \frac{\frac{1}{2}(\kappa \varepsilon_0 A/d)(Ed)^2}{Ad} \implies \mathbf{u_E} = \frac{\mathbf{1}}{\mathbf{2}} \kappa \varepsilon_0 \mathbf{E^2} \tag{35}$$

5. Attractive Mechanical Force Between Plates:

$$\mathbf{F} = -\frac{\mathbf{dU}}{\mathbf{dx}} = \frac{\mathbf{Q^2}}{\mathbf{2\varepsilon_0 A}} = \frac{\mathbf{1}}{\mathbf{2}} \mathbf{QE_0} = \frac{\mathbf{1}}{\mathbf{2}} \varepsilon_0 \mathbf{AE_0^2} \tag{36}$$

Dielectric Insertion: Battery Connected vs. Battery Disconnected

Physical Parameter	Case A: Battery Disconnected (Q = const)	Case B: Battery Remains Connected (V = const)
Charge (Q)	Unchanged: $Q = Q_0$	Increases: $Q = \kappa Q_0$
Capacitance (C)	Increases: $C = \kappa C_0$	Increases: $C = \kappa C_0$
Potential Difference (V)	Decreases: $V = V_0/\kappa$	Unchanged: $V = V_0$
Electric Field (E)	Decreases: $E = E_0/\kappa$	Unchanged: $E = E_0$
Stored Energy (U)	Decreases: $U = U_0/\kappa$	Increases: $U = \kappa U_0$
Battery Work Done	Zero (battery isolated)	$W_{\text{battery}} = \Delta Q \cdot V_0 = (\kappa - 1) C_0 V_0^2$

4 Current Electricity, Drift Dynamics & DC Resistive Networks

Drift Velocity, Microscopic Ohm's Law & Mobility

1. Microscopic Mechanism of Conduction in Metals: Under zero external field, conduction electrons execute chaotic thermal motion at speeds $v_{\text{th}} \sim 10^5\,\text{m/s}$ with random directions ($\langle \mathbf{v}_{\text{th}} \rangle = 0$). When an external electric field $\mathbf{E}$ is applied, each electron experiences an electrostatic force $\mathbf{F} = -e\mathbf{E}$, imparting constant acceleration $\mathbf{a} = -\frac{e\mathbf{E}}{m_e}$. Electrons repeatedly scatter elastically from vibrating crystal lattice ions. The average time elapsed between consecutive scattering collisions is the **relaxation time** $\tau \sim 10^{-14}\,\text{s}$. **2. Drift Velocity ($\mathbf{v}_d$) Derivation:**

$$\mathbf{v}_d = \langle \mathbf{v}_{\text{initial}} + \mathbf{a}t \rangle = 0 + \mathbf{a}\tau \implies \mathbf{v_d} = -\frac{\mathbf{eE}\tau}{\mathbf{m_e}} \tag{37}$$

Typically, drift speed is remarkably slow: $v_d \sim 10^{-4}\,\text{m/s} = 0.1\,\text{mm/s}$. **3. Current Density & Microscopic Ohm's Law:** Consider a conductor of cross-sectional area A with free electron density n. In time Δt, charge

passing through is $\Delta Q = neAv_d\Delta t$.

$$I = \frac{\Delta Q}{\Delta t} = \mathbf{neAv_d} \implies \mathbf{J} = \frac{\mathbf{I}}{\mathbf{A}} = \mathbf{nev_d} = \left(\frac{\mathbf{ne^2\tau}}{\mathbf{m_e}}\right)\mathbf{E} \tag{38}$$

$$\mathbf{J} = \sigma\mathbf{E} = \frac{\mathbf{1}}{\rho}\mathbf{E} \quad \text{(Vector / Microscopic Ohm's Law)} \tag{39}$$

where electrical conductivity σ and resistivity ρ:

$$\sigma = \frac{\mathbf{ne^2\tau}}{\mathbf{m_e}}, \qquad \rho = \frac{\mathbf{m_e}}{\mathbf{ne^2\tau}} \tag{40}$$

4. Electron Mobility (μ): Drift speed per unit electric field:

$$\mu = \frac{\mathbf{v_d}}{\mathbf{E}} = \frac{\mathbf{e\tau}}{\mathbf{m_e}} \quad [\mathrm{m^2 \cdot V^{-1} \cdot s^{-1}}] \tag{41}$$

Temperature Dependence, Cell Grouping & Power Transfer Theorem

1. Temperature Dependence of Resistance: For metallic conductors, increasing temperature intensifies lattice vibrations, reducing relaxation time τ:

$$R(T) = R_0[1 + \alpha(T - T_0)] \implies \alpha = \frac{\mathbf{R_T - R_0}}{\mathbf{R_0\Delta T}} \quad [^\circ\mathrm{C}^{-1} \text{ or } \mathrm{K}^{-1}] \tag{42}$$

For metals, $\alpha > 0$ (positive). For semiconductors and electrolytes, charge carrier density surges exponentially with temperature ($n(T) \propto e^{-E_g/2k_BT}$), overriding τ drops, giving $\alpha < \mathbf{0}$ (negative). **2. Terminal Potential Difference of a Real Cell ($\mathcal{E}, r$):**

- **Discharging (Supplying current I to load R):** $\mathbf{V = \mathcal{E} - Ir} = \frac{\mathcal{E}R}{R+r}$.
- **Charging (External charger driving current into cell):** $\mathbf{V = \mathcal{E} + Ir}$.
- **Open Circuit ($I = 0$):** $V = \mathcal{E}$. Short circuit ($R = 0$): $I_{\text{sc}} = \frac{\mathcal{E}}{r}, V = 0$.

3. Mixed Grouping of Cells (m rows of n series cells each): Total cells $N = n \times m$. Total EMF $= n\mathcal{E}$; total internal resistance $= \frac{nr}{m}$. Current through load R:

$$I = \frac{n\mathcal{E}}{R + \frac{nr}{m}} = \frac{mn\mathcal{E}}{mR + nr} \tag{43}$$

To maximize current, minimize the denominator $(mR + nr) = (\sqrt{mR} - \sqrt{nr})^2 + 2\sqrt{mnRr}$. The minimum occurs when $\sqrt{mR} = \sqrt{nr}$:

$$\mathbf{R = \frac{nr}{m}} \quad \text{(Load resistance equals total internal resistance)} \tag{44}$$

4. Maximum Power Transfer Theorem: Power delivered to load: $P(R) = I^2R = \frac{\mathcal{E}^2R}{(R+r)^2}$. Setting $\frac{dP}{dR} = 0$:

$$\frac{dP}{dR} = \mathcal{E}^2\frac{(R+r)^2 - 2R(R+r)}{(R+r)^4} = 0 \implies R + r - 2R = 0 \implies \mathbf{R = r} \tag{45}$$

$$\mathbf{P_{max}} = \frac{\mathcal{E}^\mathbf{2}}{\mathbf{4r}} \tag{46}$$

Kirchhoff's Laws & Measurement Bridges

1. Kirchhoff's Laws:

- **Junction Rule (KCL):** At any circuit node: $\sum I_{\text{in}} = \sum I_{\text{out}}$ (Conservation of Charge).
- **Loop Rule (KVL):** Around any closed loop: $\sum \Delta V = \sum \mathcal{E} - \sum IR = 0$ (Conservation of Energy).

2. Wheatstone Bridge Balance Derivation: A quadrilateral network of four resistors P, Q, R, S with a galvanometer connected between junctions B and D. At balance, no current traverses the galvanometer ($I_g = 0 \implies V_B = V_D$). Applying KVL to loop $ABDA$: $I_1P - I_2R = 0 \implies I_1P = I_2R$. Applying KVL to loop $BCDB$: $I_1Q - I_2S = 0 \implies I_1Q = I_2S$. Dividing the two equations yields the classical balance

condition:

$$\frac{\mathbf{P}}{\mathbf{Q}} = \frac{\mathbf{R}}{\mathbf{S}} \tag{47}$$

3. Metre Bridge (Slide-Wire Bridge): A uniform wire of length 100 cm acts as ratio arms $P = \rho_w l$ and $Q = \rho_w(100 - l)$.

$$\frac{R}{X} = \frac{l}{100 - l} \implies \mathbf{X = R\left(\frac{100 - l}{l}\right)} \tag{48}$$

Maximum experimental precision occurs at the central null point ($l \approx 50$ cm).

Syllabus Boundary Alert: Potentiometer & Carbon Resistor Color Codes Deletion

NMC REVISION WARNING FOR NEET UG: Under the revised NEET UG syllabus issued by the National Medical Commission:

- **STRICTLY DELETED:** The **Potentiometer** (principle of fall of potential $V \propto l$, comparison of EMFs of two cells $\mathcal{E}_1/\mathcal{E}_2 = l_1/l_2$, and determination of cell internal resistance $r = R(l_1/l_2 - 1)$) has been **completely deleted**.
- **STRICTLY DELETED: Color Coding of Carbon Resistors** (BBROYGBVGW mnemonic bands, tolerance multipliers) has been **completely deleted**.
- **RETAINED:** The **Metre Bridge** and **Wheatstone Bridge** are fully retained and form the exclusive focus for bridge measurements.

5 Magnetic Effects of Electric Current & Moving Charges

Biot-Savart Law & Magnetic Lorentz Force

1. Biot-Savart Law: The infinitesimal magnetic field $d\mathbf{B}$ generated by an elementary current length vector $Id\mathbf{l}$ at displacement $\mathbf{r}$:

$$\mathbf{dB} = \frac{\mu_0}{4\pi}\frac{\mathbf{I(dl \times \hat{r})}}{\mathbf{r^2}} = \frac{\mu_0}{4\pi}\frac{\mathbf{I(dl \times r)}}{\mathbf{r^3}} \tag{49}$$

where $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{T \cdot m \cdot A^{-1}}$ is vacuum permeability. **2. Magnetic Lorentz Force on a Moving Charge:**

$$\mathbf{F = q(E + v \times B)} \tag{50}$$

- Pure magnetic force: $\mathbf{F}_B = q(\mathbf{v} \times \mathbf{B})$. Since $\mathbf{F}_B \perp \mathbf{v}$, the instantaneous power is $P = \mathbf{F}_B \cdot \mathbf{v} = 0$.
- **Work Done by Magnetic Force is Identically Zero:** Magnetic forces change the direction of motion, never particle kinetic energy or speed.
- **Helical Trajectory in Uniform Field B:** When particle velocity $\mathbf{v}$ makes angle θ with $\mathbf{B}$ ($v_\perp = v\sin\theta, v_\parallel = v\cos\theta$):

$$\mathbf{r} = \frac{\mathbf{mv}_\perp}{\mathbf{qB}} = \frac{\mathbf{mv}\sin\theta}{\mathbf{qB}}, \qquad \mathbf{T} = \frac{\mathbf{2\pi m}}{\mathbf{qB}}, \qquad \text{Pitch } \mathbf{p} = \mathbf{v}_\parallel \mathbf{T} = \frac{\mathbf{2\pi mv}\cos\theta}{\mathbf{qB}} \tag{51}$$

Master Magnetic Field Formulas: Straight Wires, Loops, Solenoids & Toroids

1. Finite Straight Wire Subtending Angles θ_1, θ_2 at Distance d:

$$\mathbf{B} = \frac{\mu_0 \mathbf{I}}{4\pi \mathbf{d}}(\sin\theta_1 + \sin\theta_2) \tag{52}$$

- **Infinitely Long Straight Wire ($\theta_1 = \theta_2 = 90°$):** $\mathbf{B} = \frac{\mu_0 \mathbf{I}}{2\pi \mathbf{d}}$.
- **Semi-Infinite Wire End ($\theta_1 = 90°, \theta_2 = 0°$):** $\mathbf{B} = \frac{\mu_0 \mathbf{I}}{4\pi \mathbf{d}}$.

2. Circular Current Loop of Radius R on Symmetry Axis (x):

$$\mathbf{B(x)} = \frac{\mu_0 \mathbf{IR^2}}{\mathbf{2(R^2 + x^2)^{3/2}}}\hat{\mathbf{i}} \implies \mathbf{B}_{\text{center}} = \frac{\mu_0 \mathbf{I}}{\mathbf{2R}} \tag{53}$$

3. Ampere's Circuital Law:

$$\oint \mathbf{B} \cdot \mathbf{dl} = \mu_0 \mathbf{I}_{\text{enclosed}} \tag{54}$$

- **Ideal Long Solenoid (n turns/meter):** Interior field is axial and uniform: $\mathbf{B} = \mu_0 \mathbf{nI}$. Near ends: $B_{\text{end}} = \frac{1}{2}\mu_0 nI$.
- **Toroid of Mean Radius R and N Total Turns:** $n = \frac{N}{2\pi R} \implies \mathbf{B} = \mu_0 \mathbf{nI} = \frac{\mu_0 \mathbf{NI}}{\mathbf{2\pi R}}$.

4. Force Between Parallel Currents (Definition of Ampere):

$$\frac{\mathbf{F}}{\mathbf{L}} = \frac{\mu_0 \mathbf{I_1 I_2}}{\mathbf{2\pi d}} \tag{55}$$

Parallel currents **attract**; antiparallel currents **repel**.

Moving Coil Galvanometer Conversion & Magnetic Materials

1. Moving Coil Galvanometer (MCG): Deflecting torque balances radial restoring suspension torque: $\tau = NIAB = C\theta \implies \theta = \left(\frac{NAB}{C}\right) I$.

- **Conversion to Ammeter (Extending range to $I > I_g$):** Connect small shunt resistance S in parallel:

$$(I - I_g)S = I_g R_g \implies \mathbf{S} = \frac{\mathbf{I_g R_g}}{\mathbf{I - I_g}} \tag{56}$$

- **Conversion to Voltmeter (Extending range to V):** Connect high multiplier resistance R in series:

$$V = I_g(R_g + R) \implies \mathbf{R} = \frac{\mathbf{V}}{\mathbf{I_g}} - \mathbf{R_g} \tag{57}$$

2. Classification of Magnetic Materials:

- **Diamagnetic:** $\chi_m < 0$ (small, negative, independent of temperature). $\mu_r < 1$. Repelled by magnetic fields. Examples: Bi, Cu, H_2O, NaCl.
- **Paramagnetic:** $\chi_m > 0$ (small, positive). Follows **Curie's Law**: $\chi_m = \frac{C}{T}$. $\mu_r > 1$. Weakly attracted. Examples: Al, Na, O_2.
- **Ferromagnetic:** $\chi_m \gg 1$ (enormous, positive). Follows **Curie-Weiss Law** above Curie temperature T_c: $\chi_m = \frac{C}{T - T_c}$. Exhibits domains and hysteresis. Examples: Fe, Co, Ni.

Syllabus Boundary Alert: Cyclotron & Earth's Magnetism Deletion

NMC REVISION WARNING FOR NEET UG: Under the revised NEET UG syllabus issued by the National Medical Commission:

- **STRICTLY DELETED:** The **Cyclotron** (resonating RF frequency $\nu_c = \frac{qB}{2\pi m}$, kinetic energy $K_{\max} = \frac{q^2 B^2 R^2}{2m}$, relativistic limits) has been **completely removed**.
- **STRICTLY DELETED: Earth's Magnetic Elements** (Magnetic Declination θ, Angle of Dip / Inclination δ, Horizontal Component B_H, relation $B = \sqrt{B_H^2 + B_V^2}$) have been **completely removed**.

6 Electromagnetic Induction (EMI) & Inductive Circuits

Faraday's Law, Lenz's Law & Motional EMF

1. Magnetic Flux (Φ_B):

$$\Phi_B = \iint \mathbf{B} \cdot d\mathbf{A} = BA\cos\theta \quad [\text{Weber, Wb} = \text{T} \cdot \text{m}^2] \tag{58}$$

2. Faraday's Law of Electromagnetic Induction: Induced EMF in any closed circuit equals the negative time rate of change of magnetic flux:

$$\mathcal{E} = -\frac{\mathbf{d\Phi_B}}{\mathbf{dt}} = -\mathbf{N}\frac{\mathbf{d\Phi_B}}{\mathbf{dt}} \tag{59}$$

3. Lenz's Law (Conservation of Energy): The polarity of the induced EMF is such that it produces a current whose magnetic field opposes the original flux change that created it. **4. Motional EMF Derivations:**

- **Straight Conducting Rod of Length L Moving with Velocity $\mathbf{v} \perp \mathbf{B}$:** Lorentz force on free electrons $\mathbf{F} = -e(\mathbf{v} \times \mathbf{B})$ drives charge accumulation until internal electrostatic field cancels magnetic force: $eE = evB \implies \mathcal{E} = \mathbf{BvL}$.
- **Conducting Rod of Length L Rotating at Angular Velocity ω in Uniform B:** Integrating elemental motional EMF $d\mathcal{E} = Bv(r)dr = B(\omega r)dr$:

$$\mathcal{E} = \int_0^{\mathbf{L}} \mathbf{B}\omega\mathbf{r}\mathbf{dr} = \frac{\mathbf{1}}{\mathbf{2}}\mathbf{B}\omega\mathbf{L^2} \tag{60}$$

Self-Inductance, Mutual Inductance & Magnetic Energy Density

1. Self-Inductance (L): Total flux linkage is proportional to circulating current: $N\Phi_B = LI$.

$$\mathcal{E}_\mathbf{L} = -\mathbf{L}\frac{\mathbf{dI}}{\mathbf{dt}} \quad [\text{Henry}, \text{H} = \text{V} \cdot \text{s} \cdot \text{A}^{-1}] \tag{61}$$

For a long air-core solenoid of length l, area A, and turn density n:

$$\Phi = BA = (\mu_0 nI)A \implies N\Phi = (nl)(\mu_0 nIA) = \mu_0 n^2 AlI \implies \mathbf{L} = \boldsymbol{\mu_0}\mathbf{n^2}\mathbf{Al} = \boldsymbol{\mu_0}\frac{\mathbf{N^2}}{\mathbf{l}}\mathbf{A} \tag{62}$$

2. Mutual Inductance (M): $N_2\Phi_2 = M_{21}I_1 \implies \mathcal{E}_2 = -M\frac{dI_1}{dt}$. For coaxial solenoids:

$$\mathbf{M} = \boldsymbol{\mu_0}\mathbf{n_1}\mathbf{n_2}\mathbf{Al} = \frac{\boldsymbol{\mu_0}\mathbf{N_1}\mathbf{N_2}\mathbf{A}}{\mathbf{l}} \tag{63}$$

Coupling coefficient: $\mathbf{k} = \frac{\mathbf{M}}{\sqrt{\mathbf{L_1}\mathbf{L_2}}}$ $(0 \le k \le 1)$. **3. Energy Stored in an Inductor:**

$$\mathbf{U} = \int_0^{\mathbf{I}} \mathbf{Lidi} = \frac{\mathbf{1}}{\mathbf{2}}\mathbf{LI^2} \tag{64}$$

4. Magnetic Energy Density in a Magnetic Field:

$$\mathbf{u_B} = \frac{\mathbf{U}}{\text{Volume}} = \frac{\mathbf{B^2}}{\mathbf{2}\boldsymbol{\mu_0}} \tag{65}$$

NCERT Extraction: Eddy Currents & Practical Technological Applications

When bulk conductors are subjected to changing magnetic flux, circulating closed induced current loops (**Eddy Currents**) are established in planes perpendicular to the magnetic lines of force.

- **Electromagnetic Damping:** The metallic non-magnetic frame of a moving coil galvanometer develops opposing eddy currents as it deflects, rapidly bringing the pointer to rest at its steady reading without oscillatory hunting (dead-beat galvanometer).
- **Induction Furnaces:** High-frequency alternating magnetic fields generate massive eddy currents within metallic charges, producing Joule heat sufficient to melt metals in vacuum.
- **Magnetic Braking in Trains:** Electromagnets activated above rails induce opposing eddy currents in the rotating metallic wheels/rails, producing smooth, wear-free deceleration.
- **Mitigation in Transformers:** To minimize eddy current heating losses in transformers and motors, solid iron cores are replaced with thin, individually varnished/insulated metallic laminations stacked parallel to magnetic flux lines.

7 Alternating Currents (AC) & Resonant Circuits

Alternating Current Fundamentals: Phasors & RMS Values

A sinusoidal alternating EMF and current:

$$V(t) = V_0 \sin(\omega t), \qquad I(t) = I_0 \sin(\omega t - \phi) \tag{66}$$

1. Mean (Average) Value over Half-Cycle:

$$I_{\text{avg}} = \frac{1}{\pi/\omega} \int_0^{\pi/\omega} I_0 \sin(\omega t) dt = \frac{\mathbf{2}}{\pi} \mathbf{I_0} \approx \mathbf{0.637 I_0} \tag{67}$$

Full cycle average is strictly zero: $\langle I \rangle_{\text{full}} = 0$. **2. Root-Mean-Square (RMS / Virtual) Value:** Equating Joule heating over a full cycle $H = \int_0^T I^2 R dt = I_{\text{rms}}^2 RT$:

$$I_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T I_0^2 \sin^2(\omega t) dt} = \frac{\mathbf{I_0}}{\sqrt{\mathbf{2}}} \approx \mathbf{0.707 I_0}, \qquad \mathbf{V_{\text{rms}}} = \frac{\mathbf{V_0}}{\sqrt{\mathbf{2}}} \tag{68}$$

AC meters (hot-wire ammeters and voltmeters) always read **RMS values**.

Series LCR Circuit: Impedance, Resonance & Quality Factor

In a series LCR circuit with supply voltage $V(t) = V_0 \sin(\omega t)$:

$$V_R = IR, \qquad V_L = IX_L = I(\omega L), \qquad V_C = IX_C = I\left(\frac{1}{\omega C}\right) \tag{69}$$

Since V_L leads current by $\pi/2$ and V_C lags current by $\pi/2$:

$$V_0 = \sqrt{V_R^2 + (V_L - V_C)^2} = I_0 \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \tag{70}$$

$$\mathbf{Z} = \sqrt{\mathbf{R^2} + (\mathbf{X_L} - \mathbf{X_C})^2} = \sqrt{\mathbf{R^2} + \left(\omega \mathbf{L} - \frac{\mathbf{1}}{\omega \mathbf{C}}\right)^2}, \qquad \tan\phi = \frac{\mathbf{X_L} - \mathbf{X_C}}{\mathbf{R}} \tag{71}$$

1. Electrical Resonance Condition: When inductive reactance equals capacitive reactance:

$$X_L = X_C \implies \omega_0 L = \frac{1}{\omega_0 C} \implies \boldsymbol{\omega_0} = \frac{\mathbf{1}}{\sqrt{\mathbf{LC}}}, \quad \boldsymbol{\nu_0} = \frac{\mathbf{1}}{\mathbf{2\pi\sqrt{LC}}} \tag{72}$$

At resonance:

- Impedance is minimum and purely resistive: $\mathbf{Z}_{\text{min}} = \mathbf{R}$.
- Current amplitude is maximum and in phase with voltage: $\mathbf{I_{0,\text{max}}} = \frac{\mathbf{V_0}}{\mathbf{R}}$, $\phi = 0$.
- Power factor is unity: $\cos\phi = 1$.

2. Sharpness of Resonance & Quality Factor (Q):

$$\mathbf{Q} = \frac{\boldsymbol{\omega_0}\mathbf{L}}{\mathbf{R}} = \frac{\mathbf{1}}{\boldsymbol{\omega_0}\mathbf{CR}} = \frac{\mathbf{1}}{\mathbf{R}}\sqrt{\frac{\mathbf{L}}{\mathbf{C}}} = \frac{\boldsymbol{\omega_0}}{\mathbf{2}\boldsymbol{\Delta\omega}} = \frac{\text{Resonant Frequency}}{\text{Bandwidth}} \tag{73}$$

High Q-factor indicates sharp selectivity and low damping. **3. Average Power in AC Circuit & Power Factor:**

$$\mathbf{P}_{\text{avg}} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}} \cos\phi \tag{74}$$

where $\cos\phi = \frac{R}{Z}$ is the **power factor**.

- Pure L or pure C: $\phi = \pm\pi/2 \implies \cos\phi = 0 \implies P_{\text{avg}} = 0$ (**Wattless Current** $I_w = I_{\text{rms}} \sin\phi$).
- Pure R or LCR at resonance: $\phi = 0 \implies \cos\phi = 1 \implies P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}}$.

4. Transformers:

$$\frac{\mathbf{V_s}}{\mathbf{V_p}} = \frac{\mathbf{N_s}}{\mathbf{N_p}} = \frac{\mathbf{I_p}}{\mathbf{I_s}} = \mathbf{K} \quad \text{(Transformation Ratio)} \tag{75}$$

Efficiency: $\eta = \frac{P_{\text{output}}}{P_{\text{input}}} = \frac{V_s I_s \cos\phi_s}{V_p I_p \cos\phi_p}$.

8 Electromagnetic Waves & Maxwell's Equations

Maxwell's Displacement Current & Unified Field Equations

1. Maxwell's Displacement Current (I_d): Ampere's Circuital Law ($\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$) failed when applied to a charging capacitor boundary. Maxwell resolved this inconsistency by postulating that a time-varying electric field produces an equivalent displacement current:

$$\Phi_E = EA = \left(\frac{Q}{\varepsilon_0 A}\right) A = \frac{Q}{\varepsilon_0} \implies I_d = \varepsilon_0 \frac{d\Phi_E}{dt} = \frac{dQ}{dt} = I_c \tag{76}$$

$$\oint \mathbf{B} \cdot \mathbf{dl} = \mu_0(\mathbf{I_c} + \mathbf{I_d}) = \mu_0 \mathbf{I_c} + \mu_0 \varepsilon_0 \frac{\mathbf{d\Phi_E}}{\mathbf{dt}} \quad \text{(Ampere-Maxwell Law)} \tag{77}$$

2. Transverse Nature & Speed of EM Waves: Electric and magnetic vectors oscillate perpendicular to each other and perpendicular to the wave propagation direction:

$$\mathbf{c} = \frac{\mathbf{1}}{\sqrt{\mu_0 \varepsilon_0}} = \frac{\mathbf{E_0}}{\mathbf{B_0}} \approx \mathbf{3.00 \times 10^8}\ \text{m/s} \tag{78}$$

3. Poynting Vector & Radiation Pressure: Energy flux vector: $\mathbf{S} = \frac{\mathbf{1}}{\mu_0}(\mathbf{E} \times \mathbf{B})$. Average intensity: $I = \frac{1}{2}\varepsilon_0 c E_0^2 = \frac{E_0 B_0}{2\mu_0}$.

$$\text{Radiation Pressure: } P_{\text{rad}} = \begin{cases} \frac{I}{c} & \text{Perfectly Absorbing Surface} \\ \frac{2I}{c} & \text{Perfectly Reflecting Surface} \end{cases} \tag{79}$$

The Electromagnetic Spectrum: Master Classification

Wave Type	Wavelength (λ)	Frequency (ν)	Production Mechanism & Major Use
Radio Waves	> 0.1 m	$< 3 \times 10^9$ Hz	Rapid charge acceleration in LC antennas; Radio/TV communication
Microwaves	1 mm – 0.1 m	$10^9 - 10^{11}$ Hz	Klystron/Magnetron tubes; Radar navigation, Microwave ovens
Infrared	700 nm – 1 mm	$10^{11} - 4 \times 10^{14}$ Hz	Thermal molecular vibrations; Greenhouse effect, Remote controls
Visible Light	400 nm – 700 nm	$4 - 7.5 \times 10^{14}$ Hz	Electron transitions in outer atomic shells; Human vision
Ultraviolet	1 nm – 400 nm	$10^{15} - 10^{17}$ Hz	High-temperature inner shell transitions, sun; Sterilization, LASIK
X-Rays	0.01 nm – 10 nm	$10^{16} - 10^{20}$ Hz	Fast electrons decelerated on heavy metal target; Medical radiography
Gamma Rays	$< 10^{-3}$ nm	$> 10^{19}$ Hz	Nuclear de-excitation / nuclear isomeric transitions; Cancer radiotherapy

9 Master High-Yield Problem Archetypes & Worked Solutions

Archetype 1: Cube Resistor Network Equivalent Resistances

Problem: Twelve identical resistors, each of resistance R, form the edges of a skeleton cube. Compute equivalent resistance between: (i) diagonally opposite body corners, (ii) opposite face corners, and (iii) adjacent edge corners. **Step-by-Step Solution:**

1. **Body Diagonal (Points A and G):** Inject current I at A and extract at G. By threefold rotational symmetry around the body diagonal:
   - 3 edges leave A, each carrying $I/3$.
   - Each splits into 2 edges, each carrying $I/6$ across the middle 6 edges.
   - These recombine into 3 edges entering G, each carrying $I/3$.

Total potential drop: $V_{AG} = \left(\frac{I}{3}\right) R + \left(\frac{I}{6}\right) R + \left(\frac{I}{3}\right) R = IR\left(\frac{1}{3} + \frac{1}{6} + \frac{1}{3}\right) = \frac{5}{6} IR$.

$$\mathbf{R}_{\text{body}} = \frac{\mathbf{5}}{\mathbf{6}}\mathbf{R} \tag{80}$$

2. **Face Diagonal (Points A and C):** By mirror symmetry across the plane bisecting the face:

$$\mathbf{R}_{\text{face}} = \frac{\mathbf{3}}{\mathbf{4}}\mathbf{R} \tag{81}$$

3. **Adjacent Edge (Points A and B):**

$$\mathbf{R}_{\text{edge}} = \frac{\mathbf{7}}{\mathbf{12}}\mathbf{R} \tag{82}$$

Archetype 2: Series LCR Circuit Resonance & Power Factor Calculation

Problem: A series LCR circuit has $R = 10\,\Omega$, $L = 0.1\,\text{H}$, and $C = 100\,\mu\text{F}$ connected to an AC source $V(t) = 200\sin(100t)\,\text{V}$. Calculate: (i) Inductive reactance X_L, (ii) Capacitive reactance X_C, (iii) Total impedance Z, (iv) Peak current I_0, (v) Resonant frequency ω_0, and (vi) Quality factor Q. **Step-by-Step Solution:**

1. $\omega = 100\,\text{rad/s}$, $V_0 = 200\,\text{V}$.
2. $X_L = \omega L = 100 \times 0.1 = \mathbf{10\,\Omega}$.
3. $X_C = \frac{1}{\omega C} = \frac{1}{100 \times 100 \times 10^{-6}} = \frac{1}{10^{-2}} = \mathbf{100\,\Omega}$.
4. Total impedance Z:

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{10^2 + (10 - 100)^2} = \sqrt{100 + (-90)^2} = \sqrt{100 + 8100} = \sqrt{8200} \approx \mathbf{90.55\,\Omega} \tag{83}$$

Circuit is strongly capacitive ($X_C > X_L$), current leads voltage.

5. Peak current: $I_0 = \frac{V_0}{Z} = \frac{200}{90.55} \approx \mathbf{2.21}\,\text{A}$.
6. Resonant angular frequency ω_0:

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.1 \times 10^{-4}}} = \frac{1}{\sqrt{10^{-5}}} = \frac{1000}{\sqrt{10}} \approx \mathbf{316.2}\,\text{rad/s} \tag{84}$$

7. Quality factor Q:

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{10}\sqrt{\frac{0.1}{10^{-4}}} = \frac{1}{10}\sqrt{1000} = \frac{31.62}{10} \approx \mathbf{3.16} \tag{85}$$